

MatriSense

Comprehensive Product, Technical Architecture, and Business Strategy

Bangla-first AI-assisted maternal triage, referral coordination, and health-worker workflow for rural Bangladesh

Working positioning statement MatriSense is a Bangla-first AI-assisted maternal triage and referral platform that helps rural pregnant mothers report symptoms early and helps health workers prioritize high-risk cases for faster follow-up.		
Mother side Bangla symptom reporting, voice/text accessibility, clear risk guidance, warning signs, result history.	AI safety side LLM for extraction and explanation; rule engine for urgency; RAG for grounded guidance; validator for guardrails.	Health-worker side Prioritized cases, structured history, matched rules/evidence, referral notes, status updates, regional hospital assignment.

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1. Executive Summary

MatriSense is designed as a practical maternal health workflow, not a free-form medical chatbot. A pregnant mother reports symptoms in Bangla, the system extracts structured facts, asks short follow-up questions, runs a transparent danger-sign rule engine, retrieves guideline-grounded guidance, generates safe Bangla output, and sends high-risk cases to health workers for follow-up and referral.

Current progress: the core patient-to-health-worker flow is working. The mother can proceed through triage, the case reaches the health-worker dashboard, patient history and worker case detail exist, and status update support is present. The next phase should strengthen RAG sophistication, regional referral operations, privacy storytelling, documentation, and final-round readiness.

The key message for judges and partners

MatriSense helps mothers speak earlier and helps health workers act faster. It turns informal Bangla symptom reports into structured maternal risk cases while keeping the final medical decision human-led.

2. Product Identity and Positioning

2.1 Product category

- **Primary category:** AI-assisted maternal health triage and referral coordination platform.
- **Target setting:** rural and semi-rural Bangladesh, where mothers may face delayed care and health workers need structured information quickly.
- **Product shape:** two-sided workflow: mother-facing symptom reporting and health-worker-facing case management.
- **Safety stance:** MatriSense supports triage, communication, and referral; it does not diagnose disease, prescribe medicine, or replace doctors or health workers.

2.2 One-line pitch

Elevator pitch

MatriSense is a Bangla-first AI-assisted maternal triage and referral platform that helps rural pregnant mothers report symptoms early and helps health workers prioritize high-risk cases for faster follow-up.

2.3 What MatriSense is not

- Not an AI doctor.
- Not a medicine or dosage recommendation system.
- Not a generic chatbot that freely answers medical questions without guardrails.
- Not a full hospital navigation or ambulance routing platform in the MVP.
- Not a replacement for health workers, clinics, or doctors.

3. Problem Statement

Rural pregnant mothers in Bangladesh often face delayed care because warning symptoms may not be recognized early, symptoms may be described informally, and health workers may receive incomplete information too late. **The problem is not only a lack of medical advice; it is the absence of a structured, accessible bridge between mothers and frontline health workers.**

- Mothers may not know which symptoms are urgent, especially when symptoms seem common, such as headache, vomiting, swelling, fever, or abdominal pain.
- Symptoms are often reported in everyday Bangla, not in structured clinical language.
- Health workers may need to spend time collecting basic history before understanding urgency.
- High-risk maternal warning signs need faster escalation and clearer referral coordination.
- Rural health programs need workflow tools that support both patient guidance and health-worker prioritization.

4. Solution Overview

Solution in one paragraph

MatriSense provides a guided maternal triage workflow. A mother creates or uses a basic pregnancy profile, reports symptoms in Bangla text or voice, confirms extracted symptoms, answers short follow-up questions, and receives safe Bangla guidance. A rule engine classifies urgency as LOW, MEDIUM, or HIGH. A rule-aware RAG layer retrieves approved guidance based on the fixed decision. High-risk cases appear in a health-worker dashboard with profile, symptoms, follow-up answers, matched rules, evidence, guidance shown to the mother, referral notes, and status updates.

4.1 Core value propositions

Audience	Pain point	MatriSense value
Pregnant mothers	Uncertainty about symptoms and delayed help-seeking	Bangla-first symptom reporting, short follow-up questions, clear urgency guidance, warning signs, and read-aloud accessibility.
Families/caregivers	Difficulty deciding whether to seek urgent help	Simple action guidance that can be understood by family members supporting the mother.
Health workers	Unstructured cases and delayed information	Risk-prioritized dashboard, structured case records, matched danger signs, evidence, referral notes, and status tracking.
Clinics/NGOs/public health programs	Need for scalable early risk detection and referral coordination	Regional case management, audit trails, source-grounded guidance, and potential analytics over synthetic/demo and future consented data.

5. Current Progress and Maturity

Area	Current status	Next recommended action
Core patient triage flow	Working	Protect as stable baseline; avoid major rewrites.
Patient to health-worker handoff	Working	Polish case detail, profile fallbacks, status/audit consistency.
Patient triage history	Implemented or in progress	Ensure summary, risk, status, and result link are reliable.
Rule engine	Strong MVP layer	Harden with tests/source metadata; avoid large expansion now.
RAG	Functional v1	Upgrade from curated JSON/card retrieval to vector RAG, then graph-style retrieval.
Safety validator	Strong differentiator	Keep mandatory; update only when schemas change.
Regional referral/hospital assignment	Planned MVP enhancement	Assign to one teammate end-to-end.
Marketing/privacy/business narrative	Needs structured documentation	Frame as two-sided platform: mothers + health workers + institutions.
Demo/docs polish	Needs final-round upgrade	Build a clear /docs page, demo script, and evidence of safety tests.

6. End-to-End Workflow

Mother profile
 → Bangla text/voice symptom input
 → AI symptom extraction
 → symptom confirmation
 → 2-3 approved follow-up questions
 → case state builder
 → rule-based triage engine

- decision builder
- rule-aware RAG guidance retrieval
- LLM Bangla explanation
- safety validation
- mother result page
- health-worker dashboard
- case detail
- referral note / status update
- patient history and audit trail

6.1 Mother-facing flow

1. Mother logs in or enters the patient side of the app.
2. Mother creates or updates a pregnancy profile with trimester, gestational week, expected delivery date, last checkup date, known risk factors, emergency contact, and location/address fields where available.
3. Mother reports symptoms in Bangla text. Voice input can be converted to text, reviewed by the user, and then passed into the same triage flow.
4. The AI extractor converts Bangla input into symptom codes, severity, duration, and uncertain fields.
5. The mother confirms extracted symptoms and answers short follow-up questions.
6. The result page shows risk level, why the result was given, what to do now, what to monitor, warning signs, and a safety disclaimer.
7. The mother can later view triage history and result status.

6.2 Health-worker flow

8. Health worker opens the dashboard and sees cases sorted by urgency and status.
9. High-risk cases are visible with mother profile, original symptom input, extracted symptoms, follow-up answers, matched rules, evidence, and guidance already shown.
10. Health worker updates case status, adds referral/follow-up notes, and may assign or reassign a hospital when the regional referral feature is implemented.
11. The system records key actions in audit logs for traceability.

7. Product Modules

Module	Purpose	Current/target behavior
Mother profile	Stores pregnancy and contact context	Age, trimester, gestational week, EDD, last checkup, known risk factors, emergency contact, address/location.
Symptom input	Captures current complaint in Bangla	Text is core; voice is an accessibility layer if enabled.
AI extraction	Normalizes language into structured facts	LLM extracts symptoms only; it does not decide risk.
Confirmation/follow-up	Reduces extraction errors and collects missing facts	Fixed question bank, max 2–3 questions for MVP.
Rule engine	Clinical safety decision layer	Checks maternal danger signs and modifiers; outputs LOW/MEDIUM/HIGH.
RAG guidance	Source-grounded patient guidance	Retrieves allowed cards/chunks based on decision, symptoms, risk, and evidence tags.
LLM explanation	Simple Bangla result generation	Explains fixed decision using retrieved context only.
Safety validator	Final guardrail	Blocks diagnosis, medicine, false reassurance, risk downgrade, unsupported steps.
Mother result	Actionable guidance	Risk, reasons, steps, monitoring, warning signs, disclaimer.
Worker dashboard	Human-in-the-loop action	Prioritized list, case detail, notes, status updates.
Regional referral	Operational referral support	District/upazila filtering, hospital suggestions,

Module	Purpose	Current/target behavior
		assignment/reassignment.

8. Technical Architecture

8.1 Architecture principle

Safety-first AI architecture

LLM = Bangla understanding and explanation
 Rule engine = medical urgency decision
 RAG = source-grounded guidance retrieval
 Safety validator = final output guardrail
 Health worker = human-led follow-up and referral

8.2 System layers

Layer	Main components	Responsibility
Frontend	Next.js/React, Tailwind, dashboards, forms, result pages	Mother flow, worker dashboard, history, profile, referral UI.
API backend	Node.js, Express routes/controllers/services	Auth, patient, triage, worker, referral, hospital, speech, docs endpoints.
Database	MongoDB Atlas, Mongoose models	Users, patients, triage sessions, referral notes, hospitals, audit logs, knowledge cards or chunk metadata.
AI layer	LLM provider adapter, prompts, schemas	Symptom extraction, Bangla explanation, health-worker summary.
Triage layer	json-rules-engine, rules, decisionBuilder, caseStateBuilder	Risk decision and action constraints.
RAG layer	knowledgeCards, retriever, vector store roadmap, graph-style metadata	Guidance cards/chunks matched to rule decision.
Safety layer	pre-generation checks, post-generation validator, fallback templates	Prevent unsafe or contradictory patient output.

8.3 API map

Area	Endpoints
Auth	POST /api/auth/register, POST /api/auth/login, GET /api/auth/me, POST /api/auth/logout
Patients	POST /api/patients, GET /api/patients/me, GET /api/patients/:id, PUT /api/patients/:id
Triage	POST /api/triage/start, POST /api/triage/:sessionId/extract, POST /api/triage/:sessionId/confirm, GET /api/triage/:sessionId/follow-up, POST /api/triage/:sessionId/answers, POST /api/triage/:sessionId/run, POST /api/triage/:sessionId/explain, GET /api/triage/:sessionId/result
Worker	GET /api/worker/cases, GET /api/worker/cases/:sessionId, PUT /api/worker/cases/:sessionId/status, planned PUT /api/worker/cases/:sessionId/hospital
Referral/audit	POST /api/referral-notes, GET /api/referral-notes/:sessionId, GET /api/worker/cases/:sessionId/audit if enabled
Hospitals planned	GET /api/hospitals, GET /api/hospitals/nearby, POST /api/hospitals/seed-demo
Dev/admin tools	AI triage lab, rule debugger, RAG preview, safety test endpoints for development and demonstration.

9. Data Model Summary

Model	Key fields	Purpose
User	name, phone/email, passwordHash, role, coverageDistricts, canViewAllDistricts	Authentication, role-based access, health-worker regional coverage.
Patient	userId, age, phone, trimester, gestationalWeek, EDD, lastCheckupDate, knownRiskFactors, emergency contact, district/upazila/address/GPS	Permanent mother profile and pregnancy context.
TriageSession	patientId, profileSnapshot, inputBn, extractedSymptoms, confirmedSymptoms, followUpAnswers, caseState, decision, careGuidanceContext, safeOutput, status, assignedHospitalSnapshot	Core case record built from the triage interaction.
ReferralNote	triageSessionId, healthWorkerId, actionTaken, referredTo, followUpDate, note, statusAfterNote	Human follow-up and referral documentation.
AuditLog	action, actor, role, sessionId, patientId, metadata, timestamp	Traceability for important case events.
Hospital planned	name, type, division, district, upazila, address, lat/lng, phone, services, isActive	Seeded facility database for regional referral.
KnowledgeCard / guideline chunk	symptoms, riskLevelAllowed, guidanceType, source, steps, warning signs, doNotSay, embedding metadata	RAG evidence and patient-safe guidance source.

10. AI, Rule Engine, RAG, and Safety

10.1 LLM responsibilities

- Extract structured symptoms, severity, duration, and uncertain fields from Bangla text.
- Help phrase Bangla patient guidance from fixed decision and retrieved guidance context.
- Generate health-worker summaries from structured facts.
- Never decide final risk level, diagnose disease, prescribe medicine, or invent unsupported advice.

10.2 Rule engine responsibilities

- Detect maternal danger signs such as bleeding, convulsion, difficulty breathing, severe abdominal pain, headache with blurred vision, fever with severe weakness, fainting, and reduced fetal movement in later pregnancy.
- Apply modifiers such as hypertension, anemia, diabetes, previous high-risk pregnancy, late pregnancy, and long gap since checkup.
- Produce a decision package: riskLevel, priority, recommendedAction, matchedRules, reasons, evidenceTags, allowedGuidanceType, and LLM constraints.
- Prevent risk downgrade by later AI generation steps.

10.3 Current RAG v1

Current RAG design: rule-aware knowledge-card retrieval. The rule engine first determines risk and allowed guidance type. The RAG layer retrieves approved maternal guidance cards based on symptoms, evidence tags, riskLevelAllowed, and guidanceType. The LLM receives only the fixed decision and retrieved guidance.

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Decision package
  riskLevel: HIGH | MEDIUM | LOW
  evidenceTags: [headache_vision_danger]
  allowedGuidanceType: URGENT_ESCALATION_ONLY
  ↓
Rule-aware retriever
  filters by symptoms + riskLevelAllowed + guidanceType + evidenceTags
  ↓
careGuidanceContext
  stepsNowBn, monitorBn, urgentWarningBn, sources, blockedAdvice
  ↓

```

LLM explanation + safety validator

10.4 Vector RAG roadmap

Stage	Goal	Implementation idea
Vector RAG v1	Upgrade retrieval beyond JSON filtering	Embed guideline chunks or knowledge cards; store in MongoDB Atlas Vector Search or another vector store; retrieve top-k chunks with metadata filters.
Hybrid retrieval	Improve recall and precision	Combine keyword matching for symptom/evidence tags with vector search for semantic similarity.
Reranking	Prioritize source relevance	Boost cards/chunks matching evidenceTags, matchedRules, riskLevel, and allowedGuidanceType.
RAG debug panel	Make retrieval explainable	Show query, filters, retrieved chunks/cards, sources, and blocked advice in admin/dev UI.
Graph-style RAG	Add explainable relations	Represent symptom → danger sign → rule → risk level → action → source relationships.
Full GraphRAG later	More advanced final-round architecture	Use knowledge graph traversal plus vector retrieval while keeping rule engine as the risk authority.

10.5 Safety validator

- Pre-generation validation checks whether the decision and retrieved context are internally consistent.
- Post-generation validation checks the generated Bangla output before the mother sees it.
- Blocks diagnosis, medicine/dosage advice, certainty claims, false reassurance, risk downgrade, unsupported home remedies, and HIGH-risk home-care-first guidance.
- Uses fallback templates when LLM output is unsafe or missing required fields.
- Keeps MatriSense as a triage and referral support tool rather than a diagnosis engine.

11. Data Lifecycle, Privacy, and Trust

11.1 Data collected

Data type	Purpose	Privacy note
Profile data	Pregnancy context and follow-up	Collect only fields needed for triage/referral; keep profile editable.
Symptom reports	Triage case creation	Stored as part of case record; shown to authorized health workers.
Follow-up answers	Risk clarification	Stored in same triage session for traceability.
AI/rule/RAG outputs	Result, evidence, and auditability	Store decision, matched rules, retrieved sources, and safe output.
Location/GPS optional	Regional filtering and hospital referral	Ask permission; allow manual address; do not block emergency triage if denied.
Referral notes/status	Human follow-up tracking	Visible to authorized health workers and admins.
Audit logs	Accountability and troubleshooting	Record important events without exposing unnecessary data.

11.2 Privacy-by-design principles

- Use synthetic/demo data for competition demonstrations; do not claim real patient deployment data.

- Use role-based access: mothers see their own cases; health workers see assigned/authorized regional cases; admins have controlled access.
- Make GPS consent explicit and optional. If denied, continue with manual district/address or emergency triage.
- Store environment secrets outside source code and keep API keys server-side.
- Use HTTPS in deployment and secure database access controls.
- Minimize patient-identifiable information in demo screenshots and public materials.
- Maintain audit logs for case access, status updates, referral notes, and hospital assignments.
- Clearly state that MatriSense does not diagnose, prescribe medicine, or replace healthcare professionals.

12. Regional Referral and Hospital Assignment

Purpose: make MatriSense feel like an operational referral coordination system. Health workers should not see a flat list of all cases; they should be able to view cases by region, see patient address/location, and assign an appropriate facility manually.

12.1 MVP scope

- District/upazila fields in patient profile and triage session snapshot.
- Health-worker coverageDistricts and optional canViewAllDistricts flag.
- Worker case filtering by district, riskLevel, status, and assignedHospitalId.
- Seeded hospital database with district, type, services, phone, and coordinates where available.
- Nearby/same-district hospital lookup using Haversine distance when coordinates exist, or district match when they do not.
- Manual hospital assignment/reassignment by health worker with reason and audit log.
- Optional Leaflet map after list-based assignment works.

12.2 What to avoid in MVP

- Automatic hospital assignment by AI.
- Automatic health-worker assignment or complex load balancing.
- Live dependency on external hospital APIs during judging.
- Full navigation, route optimization, or ambulance dispatch.
- Blocking emergency triage because location is missing.

13. Go-to-Market and Business Strategy

13.1 Positioning

Market positioning

MatriSense should be marketed as a maternal triage and referral coordination platform, not only as a mother-facing symptom app. The two-sided value is central: mothers can report symptoms early, and health workers receive structured, prioritized cases for faster follow-up.

13.2 Target segments

Segment	Primary message	Adoption route
Pregnant mothers	Report symptoms in Bangla and understand when to seek help.	Clinic onboarding, community health sessions, family awareness, Bangla-first mobile/web flow.
Families/caregivers	Help mothers act faster when warning signs appear.	Awareness content, shared result guidance, emergency contact workflow.
Health workers	Receive structured high-risk cases instead of unorganized symptom calls.	Training, dashboard demo, regional case filtering, referral workflow.
Clinics/hospitals	Improve triage intake and referral readiness.	Pilot with antenatal care units, local maternal programs, facility dashboards.
NGOs/public health	Deploy a scalable maternal risk-detection and	B2B2C partnerships, district pilots, analytics and reporting

Segment	Primary message	Adoption route
programs	referral layer.	roadmap.

13.3 Business model options

- **B2B2C / institutional deployment:** mothers use the app for free or low cost; clinics, NGOs, or health programs pay for dashboards, referral tracking, deployment, and support.
- **Pilot partnership model:** run district-level pilots with clinics or maternal health NGOs using synthetic/demo data first, then consented deployment data later.
- **Public health program integration:** position MatriSense as a digital triage layer for community health workers and maternal care networks.
- **CSR/social impact sponsorship:** rural maternal health access can be funded by corporate or donor-backed health initiatives.

13.4 Marketing message hierarchy

Level	Message
Emotional	Helping mothers speak sooner, helping health workers act faster.
Operational	From Bangla symptom reports to prioritized maternal referral cases.
Technical	AI extraction + rule-based triage + rule-aware RAG + safety validation + human-in-the-loop referral.
Safety	MatriSense does not replace doctors; it structures early warning signs and supports faster referral.

14. Competition Strategy

14.1 Preliminary round story

- Show one complete case in under three minutes: mother profile → Bangla symptoms → AI extraction → follow-up → risk result → health-worker dashboard → status/referral note.
- Emphasize that MatriSense is not a chatbot; it is a structured maternal triage workflow.
- Demonstrate a high-risk case such as severe headache with blurred vision and show that it reaches the worker dashboard.
- Mention synthetic data, privacy boundaries, and human-led final decision clearly.

14.2 Final-round sophistication plan

- Vector RAG with metadata filtering and source/chunk citations.
- Graph-style symptom-danger-risk-action-source representation.
- Regional referral and hospital assignment with district filtering and optional map.
- Stronger /docs page with architecture, safety, data lifecycle, demo credentials, and limitations.
- Evaluation set with 25–30 synthetic triage test cases and RAG retrieval checks.
- Privacy and trust narrative: role-based access, optional GPS, audit logs, synthetic demo data, no diagnosis/prescription.

15. Success Metrics and Evaluation

Category	Metric or evidence
Functional demo	A complete mother-to-health-worker case can be demonstrated reliably.
Safety	High-risk rules never downgrade; HIGH results never show home-care-first advice.
Extraction quality	Bangla symptom extraction works for common synonyms and danger signs.
RAG quality	Retrieved guidance matches symptoms, riskLevelAllowed, guidanceType, and evidenceTags.
Workflow value	Health worker sees original input, follow-up answers, matched rules, guidance shown, notes, and status.

Category	Metric or evidence
Privacy	Demo uses synthetic data; GPS optional; role-based access and audit logs visible.
Business readiness	Clear target segments, deployment model, institutional value, and pilot plan.

16. Risks and Mitigations

Risk	Impact	Mitigation
LLM extraction error	Wrong symptoms could affect triage flow	Confirmation screen, keyword fallback extractor, approved symptom code list.
Unsafe LLM advice	Medical harm or credibility loss	Rule engine controls risk, RAG restricts guidance, safety validator and fallbacks.
Overbuilding new features	Breaks working MVP	Protect stable patient→triage→worker flow; use feature branches.
Regional referral complexity	Map/GPS consumes too much time	Complete district/hospital list assignment first; map optional.
Privacy concern	Mentor/judge concern and trust issue	Explicit consent, role-based access, synthetic demo data, audit logs, privacy section.
RAG overclaiming	Loss of credibility	Truthfully separate current JSON/card RAG from planned vector/GraphRAG.
Rule expansion burden	Testing complexity and clinical risk	Harden current rules with tests and metadata; avoid large expansion now.

17. Team Workstreams After Deadline Extension

Workstream	Owner	Main tasks
Advanced RAG	AI/RAG lead	Vector store, metadata filters, hybrid retrieval, RAG debug, graph-style metadata, retrieval tests.
Regional referral	Regional referral owner	Location fields, worker coverage, hospital model/seed, hospital lookup, assignment, audit, optional map.
Business/marketing/privacy	Business analysis owner	Market positioning, target segments, go-to-market, business model, privacy/trust document, pitch sections.
Product polish	Shared	Profile fallback, patient history polish, worker case detail, /docs page, demo script, screenshots.
Testing and stabilization	Shared	End-to-end demo cases, safety tests, RAG retrieval tests, role routing, deployment check.

18. Recommended Immediate Next Steps

- Commit and protect the working patient → triage → health-worker flow as a stable baseline.
- Assign one teammate full ownership of regional referral and hospital assignment using the prepared feature spec and prompt array.
- Continue RAG upgrade work without touching worker regional files where possible.
- Create a concise privacy/trust section for /docs and presentation.
- Polish health-worker case detail and patient history so no screen shows blank or broken profile data.
- Build a final demo script around one low-risk, one medium-risk, and one high-risk case.
- Prepare a strong BuildFest narrative: two-sided maternal triage and referral platform, not just a chatbot.

19. Demo Scenarios

Scenario	Input/profile	Expected result	Demo purpose
Low risk	Second trimester; mild nausea and mild headache; no blurred vision/swelling; can keep water down.	LOW; self-care + monitor + warning signs.	Shows safe guidance without over-escalation.
Medium risk	Fever or vomiting with weakness but no severe danger signs.	MEDIUM; contact health worker/clinic soon; monitor escalation triggers.	Shows middle path and follow-up value.
High risk	Third trimester, hypertension history; “আমার মাথা খুব ব্যথা করছে আর চোখে ঝাপসা দেখছি”.	HIGH; urgent clinic/health-worker contact; appears on dashboard.	Shows core danger-sign triage and human handoff.
Regional referral	High-risk case with district/address saved.	Worker filters by district, sees hospitals, assigns/reassigns facility.	Shows production-readiness and workflow value.

20. Source Materials Consolidated

This document consolidates and updates the team’s previous planning materials, including:

- Context summary for competition planning and project direction.
- Justification notes explaining how MatriSense builds structured maternal health records from symptom interactions.
- 7-day MVP task distribution and vertical-slice plan.
- Updated RAG implementation plan and submission-field strategy.
- Revised module ownership plan.
- Regional referral and hospital assignment feature specification and prompt array.
- Current working knowledge from the latest implementation discussions: patient → triage → health-worker flow works; patient history, worker case detail, and status update exist; regional referral and advanced RAG are next major workstreams.

Appendix A — Submission-Ready Text Blocks

A.1 Public summary

MatriSense is a Bangla-first AI-assisted maternal triage and referral platform for rural Bangladesh. Pregnant mothers can report symptoms in Bangla, confirm what the system understood, answer short follow-up questions, and receive safe guidance. The system uses an LLM for Bangla understanding and explanation, a rule engine for maternal danger-sign triage, rule-aware RAG for source-grounded guidance, and a safety validator to block diagnosis, medicine advice, false reassurance, and unsupported recommendations. High-risk cases are sent to a health-worker dashboard with structured case records, matched rules, retrieved evidence, patient-facing guidance, referral notes, and status tracking.

A.2 Problem statement

Rural pregnant mothers often face delayed care because warning symptoms are not recognized early, symptoms are not reported clearly, and frontline health workers may not receive structured case information early enough. MatriSense addresses this gap by turning Bangla symptom reports into structured maternal risk cases, checking danger signs early, giving safe urgency guidance, and sending high-risk cases to health workers for faster follow-up and referral.

A.3 Solution description

A mother reports symptoms in Bangla text or reviewed voice transcript. The AI extracts structured symptoms, severity, duration, and uncertain fields. The system asks short approved follow-up questions and builds a triage case state. A transparent rule engine classifies urgency as low, medium, or high and controls the type of guidance allowed. Rule-aware RAG retrieves trusted guidance cards or chunks matched to the symptoms, risk level, matched rules, and allowed guidance type. The LLM generates simple Bangla guidance using only the fixed decision and retrieved context, and the safety validator checks the final output. High-risk cases appear in the health-worker dashboard for follow-up, referral notes, status update, and future regional hospital assignment.

Appendix B — Glossary

Term	Meaning in MatriSense
Triage	Urgency classification and next-step guidance, not diagnosis.
Danger sign	A symptom or symptom combination that may require urgent attention during pregnancy.
Case state	Structured object containing profile, symptoms, severity, duration, follow-up answers, and risk factors.
Rule engine	Deterministic logic layer that decides LOW/MEDIUM/HIGH risk.
RAG	Retrieval-Augmented Generation: retrieving trusted guidance before LLM explanation.
Rule-aware RAG	RAG filtered by rule decision, symptoms, risk level, evidence tags, and allowed guidance type.
Safety validator	Checks AI output for unsafe medical advice before showing it to the mother.
Profile snapshot	Copy of relevant patient/profile/location fields stored on the triage session at the time of reporting.
Regional referral	District/upazila case filtering and manual hospital assignment by health workers.